

Question 4: Experimental Analysis of CNN and RNN Models for Accelerometer Sequence Classification

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1 Introduction

In this report, we analyze the performance of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for accelerometer sequence classification.

We conduct controlled experiments by varying:

- Convolutional kernel sizes
- RNN hidden state dimensions
- Number of training epochs

Training and validation accuracy are analyzed to understand model capacity, generalization, and overfitting behavior.

2 (a) Effect of Kernel Size in CNN

We varied the kernel sizes of the three convolutional layers while keeping all other hyperparameters fixed.

Kernel configurations tested:

$(3, 3, 3), (5, 3, 3), (5, 5, 3), (7, 5, 3), (9, 7, 5), (15, 11, 7), (21, 15, 9)$

Results

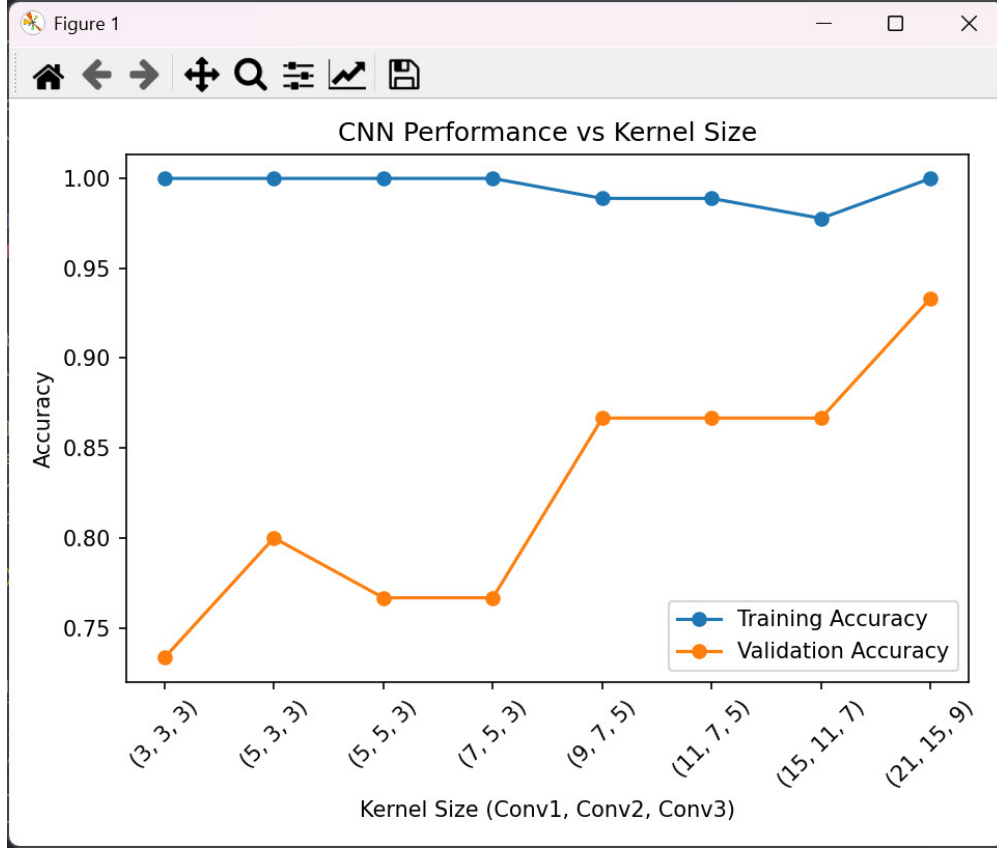


Figure 1: Training and Validation Accuracy vs Kernel Size

Discussion

Training accuracy remains near 100% across configurations, indicating that the CNN has sufficient capacity to fit the training data even with smaller kernels.

Validation accuracy generally increases with kernel size and achieves its highest value at configuration (21, 15, 9).

This suggests that larger kernels allow the model to capture longer temporal dependencies in the accelerometer signal. However, extremely large kernels may increase computational cost and risk overfitting, even though validation accuracy improved within the tested range.

Increasing kernel size increases model capacity, reducing bias but potentially increasing variance.

3 (b) Effect of Hidden State Size in RNN

Hidden sizes tested:

32, 64, 128, 256, 512

Results



Figure 2: Training and Validation Accuracy vs Hidden Size

Discussion

Training accuracy increases steadily as hidden size grows, reaching nearly 100% for larger hidden states.

Validation accuracy peaks at hidden size 128, then decreases at 256, and partially recovers at 512 but does not exceed the performance at 128.

This indicates that increasing hidden size improves representational capacity, but excessively large hidden states introduce overfitting effects.

Hidden size 256 provides the best balance between capacity and generalization among tested values.

4 (c) Effect of Number of Epochs

CNN Results

Epochs tested: 1, 5, 10, 15, 20

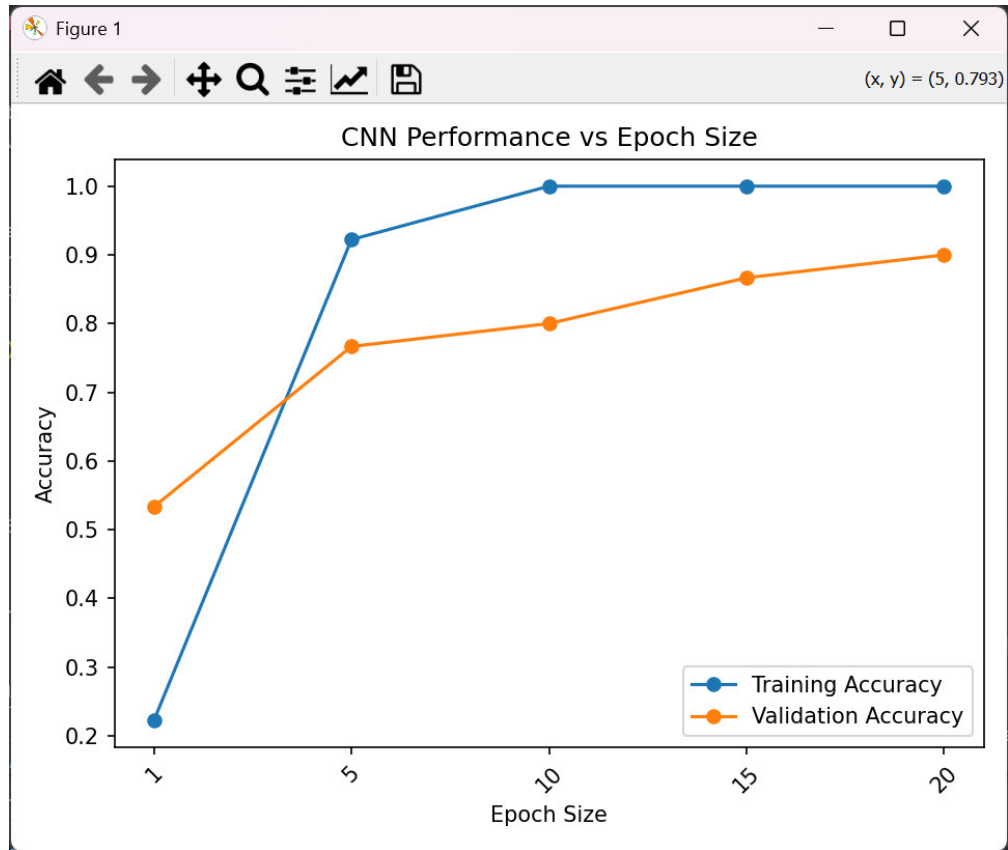


Figure 3: CNN Training and Validation Accuracy vs Epoch

Validation accuracy increases steadily across the tested epochs, while training accuracy quickly reaches 100%.

Although validation continues to improve up to 20 epochs, the large generalization gap (training = 100%) suggests overfitting risk as training progresses.

RNN Results

Epochs tested: 1, 5, 10, 15, 20, 25, 30, 35

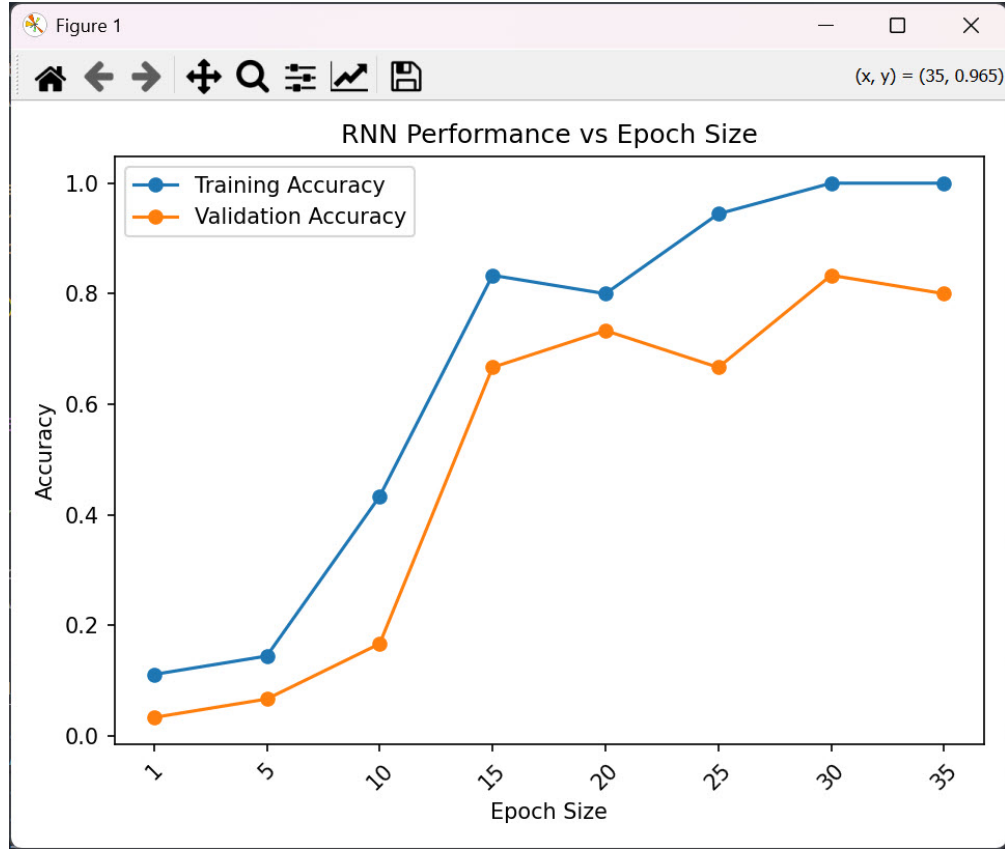


Figure 4: RNN Training and Validation Accuracy vs Epoch

Validation accuracy increases steadily and peaks around 30 epochs, after which it slightly decreases while training accuracy continues to increase.

This behavior demonstrates classic overfitting: the generalization gap widens as training continues beyond the optimal epoch count.

5 (d) Best Performing Models

Best CNN Model

- Kernel configuration: (21, 15, 9)
- Epochs: 10
- Training Accuracy: 100%
- Validation Accuracy: 87%
- Test Accuracy: 85%

Best RNN Model

- Hidden size: 256
- Epochs: 30
- Training Accuracy: 100%
- Validation Accuracy: 87%
- Test Accuracy: 80%

Model	Train Acc	Val Acc	Test Acc
Best CNN	100%	87%	85%
Best RNN	100%	87%	80%

Table 1: Final Model Performance Comparison

Comparison

The CNN and RNN achieved about the same validation accuracy.

However, the CNN achieved a slightly higher test accuracy (85%) than the RNN (80%), suggesting that the higher validation accuracy did not fully translate into improved generalization on the unseen test set.

Both architectures exhibit strong capacity to fit training data, but generalization remains limited by overfitting.

6 Conclusion

From these experiments, we conclude:

- Larger kernel sizes improve CNN validation performance within the tested range.
- Hidden size 256 provides the best RNN generalization.
- Both CNN and RNN models overfit as training progresses.
- Validation performance differences do not necessarily translate to improved test accuracy.

These results highlight the importance of hyperparameter tuning and monitoring generalization gap in time-series classification.